

Report
Of
Topic

Submitted

To

School of Computer Science and Engineering
IILM University, Greater Noida, U.P.

In partial fulfilment of the requirement of degree of
B.Tech CSE



By

Roll Number of Student: 25SCS1003000866

Name of Student: (All Capital) AMITAVA MONDAL

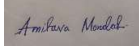
Batch: 2025-2029

2026-27

CANDIDATE'S DECLARATION

I, ___Amitava Mondal___ do hereby declare that the internship report titled “**Topic**” has been completed by me to fulfil the requirement for the award of the degree of Bachelor of Technology in CSE. All the references have been quoted and I have not taken as such any material from any other source. I affirm to you that this report is my own and has not been submitted to any other institute for any degree/diploma requirement and further I shall be solely responsible for any kind of copyright violation in this regard.

Signature :



Student Name _Amitava_ Mondal_____

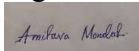
Roll No. _25SCS1003000866_____

ACKNOWLEDGEMENT

I am using this opportunity to express my gratitude to everyone who supported me throughout the Internship Program. I am thankful for their aspiring guidance, invaluable constructive criticism and friendly advice during this work. I am sincerely grateful to them for sharing their truthful and illuminating views on a number of issues related to this work.

I express my gratitude to my parents for their blessings.

Signature :

A small rectangular box containing a handwritten signature in cursive script, which appears to read "Amitava Mondal".

Student Name_Amitava_Mondal_____

Roll No.._25SCS1003000866__

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CERTIFICATE OF COMPLETION

JUN, 2026

AMITAVA MONDAL

Generative Ai Engineer

Duration: April 12, 2026, to June 12, 2026

This is to certify that Mr. Amitava Mondal has successfully completed his internship at Digital Hercules Innovations from April 12, 2026, to June 12, 2026.

We found Amitava to be skilled, detail-oriented, and proactive in his work. We recommend him for future opportunities in App development and Website technology domains.



CEO, TARANPREET
SINGH KALRA

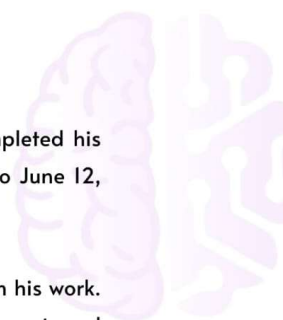
PARIKSHIT JADHAV

PROJECT MANAGER,
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AUTHORISING
AUTHORITY

Certificate ID: DHI-AI-0001-31-06-26



Introduction

I completed my internship as a Gen AI Engineer Intern at Digital Hercules Innovations (CreateAiGenie), a B2B SaaS organization working on Amazon Ads optimization. During the internship, I worked with the AI/ML team on a real-world advertising optimization platform, gaining practical exposure to the integration of APIs, cloud services, data pipelines, machine learning, and Generative AI.

The primary focus of my work was on API handling and integration scenarios within the platform. I worked on understanding how advertising-related data is received through APIs, processed by backend services, and transferred between different components of the system. This included understanding secure API communication, request and response handling, authentication mechanisms, and the role of APIs in connecting external advertising services with the application's backend.

Alongside API-related work, I contributed to the AI/ML workflow by working with advertising-performance data, including dataset preprocessing and feature engineering for ML-based ad-spend rule automation. This helped me understand how raw advertising data can be transformed into useful inputs for machine-learning and optimization workflows.

The internship provided me with practical experience in applying concepts from Generative AI, Machine Learning, backend development, API integration, and cloud computing to a production-oriented business problem. It also helped me understand how different components such as data ingestion, APIs, ML models, rule engines, and AI-based reasoning can work together to support automated and intelligent Amazon Ads optimization.

Organization Profile

Digital Hercules Innovations Pvt. Ltd. (CreateAiGenie) is a technology-focused organization based in Noida, Uttar Pradesh, operating in the areas of e-commerce, advertising technology, artificial intelligence, and SaaS solutions. The company focuses on using AI and automation to improve the efficiency and scalability of digital advertising for businesses.

Its flagship platform, CreatAiGenie, is an AI-powered Amazon PPC optimization platform designed to help Amazon sellers manage and optimize their advertising campaigns. The platform provides capabilities such as automated bid optimization, keyword harvesting, negative targeting, campaign analytics, predictive insights, and rule-based automation. It integrates with Amazon Ads and Selling Partner APIs to exchange advertising and seller data. The organization aims to reduce manual effort involved in managing Amazon advertising campaigns while helping sellers improve advertising efficiency and profitability. CreatAiGenie combines AI, machine learning, automation, and API-based integrations to analyze campaign performance and support data-driven advertising decisions.

From a technical perspective, the organization works with technologies including Python, RESTful APIs, cloud platforms, Docker, databases, OAuth/JWT authentication, and AI/ML technologies. This environment provided me with practical exposure to API integration, cloud-based systems, data processing, and AI-driven optimization during my internship.

My internship was conducted within this technical environment, where I contributed to API handling and integration scenarios and supported AI/ML workflows involving advertising-performance data, preprocessing, and feature engineering.

Problem Statement

Managing and optimizing Amazon advertising campaigns involves handling large volumes of campaign and ad-performance data from multiple sources. Manually analyzing this data and making optimization decisions can be time-consuming and difficult to scale. The platform therefore requires reliable API-based data integration and automated processing to collect, transfer, and organize advertising information efficiently.

Another challenge is converting the collected advertising data into meaningful insights that can support decisions such as bid optimization, campaign diagnosis, and ad-spend management. This requires proper data preprocessing, feature engineering, rule-based analysis, and machine-learning workflows.

During my internship, I worked primarily on API handling and integration scenarios and supported the AI/ML workflow involving advertising-performance data. The problem addressed through this work was therefore to establish an efficient flow from advertising data ingestion and API communication to data processing and AI/ML-based optimization, while maintaining secure and reliable communication between the different components of the platform.

Project Objectives

The main objectives of the project undertaken during my internship were:

1. To understand and implement API handling for communication between different components of the Amazon Ads optimization platform.
2. To establish reliable data flow between external advertising services, cloud-based services, and the application backend.
3. To work with advertising-performance data and understand its role in campaign optimization and decision-making.
4. To perform data preprocessing and feature engineering to prepare advertising data for machine-learning workflows.
5. To support ML-based ad-spend rule automation by working with model prototypes and evaluating their usefulness for advertising optimization.
6. To understand the integration of AI and automation with APIs and backend services for reducing manual effort in Amazon Ads management.
7. To gain practical experience in developing and working with production-oriented AI/ML systems and cloud-based application workflows.

Scope of the Project

The scope of the project covers the use of **API integration, data processing, machine learning, and Generative AI** within an Amazon Ads optimization platform. The project focuses on creating an efficient flow of advertising data from external services to the backend system, where it can be processed and used for optimization.

The major areas covered in the project are:

1. **API Integration and Handling** – Handling API requests and responses and understanding communication between external services and the application backend.
2. **Advertising Data Ingestion** – Working with advertising-performance data received through cloud-based data pipelines and APIs.
3. **Data Preprocessing** – Cleaning, transforming, and preparing advertising data for further analysis and machine-learning workflows.
4. **Feature Engineering** – Identifying and preparing relevant features from advertising data to support ML-based analysis.
5. **ML-Based Optimization** – Supporting model prototypes and rule automation for optimizing advertising spend and campaign performance.
6. **Generative AI Integration** – Exploring the use of LLM-based reasoning to interpret campaign data and generate optimization recommendations.
7. **Cloud and Backend Integration** – Understanding how services such as APIs, cloud infrastructure, and backend components work together to create a scalable advertising optimization workflow.

Overall, the project focuses on **automating and improving the process of collecting, processing, analyzing, and utilizing Amazon advertising data**, while reducing the need for repetitive manual analysis

Technologies and Tools Used

The project involved a combination of **programming, API integration, cloud computing, data processing, machine learning, and Generative AI** technologies.

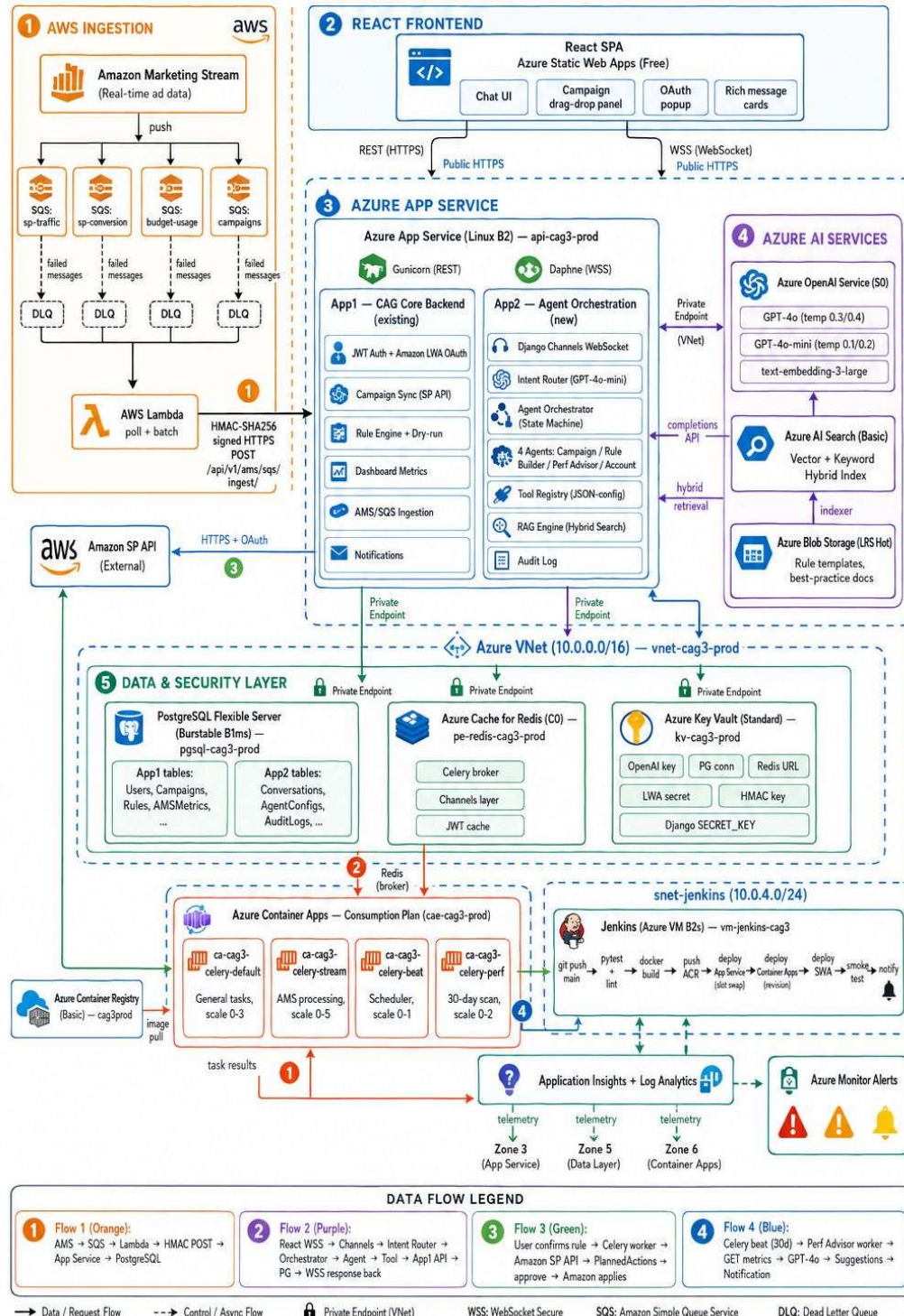
1. **Python** – Used for data processing, preprocessing, feature engineering, and machine-learning workflows.
2. **Django** – Used as part of the backend application for handling APIs and server-side operations.
3. **REST APIs** – Used for communication between the application backend and external services, enabling structured request and response handling.
4. **Amazon Web Services (AWS)** – Used for cloud-based components of the platform, particularly **AWS SQS** for message queuing and **AWS Lambda** for processing incoming data.
5. **Amazon Ads APIs** – Used for interacting with Amazon advertising-related data and services.
6. **OAuth and JWT** – Used for authentication and secure access to APIs and application services.
7. **HMAC-SHA256** – Used for secure request signing in API communication.
8. **Machine Learning** – Used for analyzing advertising-performance data and supporting ad-spend rule automation.
9. **Generative AI / LLMs** – Used for exploring intelligent reasoning and generating insights from campaign-related information.
10. **Jupyter Notebook** – Used for experimentation, data analysis, preprocessing, and evaluation of ML workflows.
11. **Git and GitHub** – Used for version control and managing development work.

These technologies supported the overall workflow of **API-based data ingestion → data preprocessing → feature engineering → ML analysis → advertising optimization**. My internship work particularly involved API handling and the AI/ML workflow around advertising-performance data.

System Architecture

CAG Agentic Platform — Production Cloud Architecture

Azure (Central India) + AWS Cross-Cloud | Terraform-managed | Jenkins CI/CD



Methodology

The methodology adopted during the internship focused on understanding and working with the **API and data-processing workflow** of the Amazon Ads optimization platform. The process began with understanding the flow of advertising data and the communication between different system components through APIs.

The major stages of the methodology were:

1. **Understanding the System Workflow** – Studying how advertising data moves between external services, cloud components, APIs, and the backend application.
2. **API Handling** – Working with API requests and responses and understanding authentication, validation, and secure communication mechanisms.
3. **Data Ingestion and Processing** – Understanding how incoming advertising-performance data is received and prepared for further processing.
4. **Data Preprocessing** – Cleaning and transforming the collected data into a suitable format for analysis.
5. **Feature Engineering** – Identifying and preparing relevant features from the advertising data for machine-learning workflows.
6. **ML Workflow Support** – Supporting model prototypes and rule-based approaches for automating advertising-spend optimization.
7. **Testing and Evaluation** – Testing API flows and data-processing workflows to identify issues and improve reliability.

The overall methodology can be represented as:

API Handling → Data Ingestion → Data Preprocessing → Feature Engineering → ML/Rule Processing → Evaluation

This methodology provided practical experience in connecting **API-based systems with data and machine-learning workflows** for a real-world Amazon Ads optimization platform.

Expected Outcomes

The project was expected to achieve the following outcomes:

1. **Reliable API Integration** – Develop a better understanding of handling API requests and responses and establishing communication between different components of the platform.
2. **Efficient Data Flow** – Enable advertising-performance data to move efficiently from data-ingestion components to the backend for further processing.
3. **Improved Data Preparation** – Prepare clean and structured datasets through preprocessing and feature engineering for ML workflows.
4. **Support for Ad Optimization** – Contribute to ML model prototypes and rule-based automation aimed at improving advertising-spend optimization.
5. **AI-Driven Insights** – Support the use of AI/ML techniques to analyze campaign performance and assist in generating useful optimization insights.
6. **Reduced Manual Effort** – Contribute to automating repetitive data-processing and advertising-optimization activities.
7. **Practical Technical Experience** – Gain hands-on experience in **API integration, cloud services, data pipelines, machine learning, and Generative AI** through work on a real-world B2B SaaS platform.

Communication mail proof



DIGITAL HERCULES INNOVATIONS PRIVATE LIMITED

Gaur city-2, Greater Noida, Uttar Pradesh, India, 201009

Date: 12-04-26

Dear **Amitava Mondal**,

We are pleased to offer you the position of **Gen Ai Engineer Intern** at Digital Hercules Innovations. We believe your skills will add great value to our team.

Internship Type: Unpaid, Experience-Based

Duration: 2 Months

Mode: Hybrid

Location: Amity Innovation Incubator (ATPC), Sector 125, Noida & IILM Innovation Incubator, Greater Noida / Gurgaon – CSE Batch (Hybrid work model)

Conversion Opportunity: Yes (Paid Internship extension / Full-time, subject to performance & funding)

Department: Engineering

Reporting To: Chief Technology Officer (CTO) / Founders

About CreateAiGenie

CreateAiGenie is a B2B SaaS platform focused on ads optimization using rule engines and ML, starting with Amazon Ads, and expanding to Google and Meta Ads.

The platform includes real dashboards, rule engines, ML-powered recommendations, and automation workflows, all built for production use. Frontend interns work on live features, not demo projects.

Role Overview

As an Intern, you will Support the AI/ML team in working with real ad-performance data, experiments, and model prototypes for Amazon ads optimisation.

You will collaborate closely with backend and product teams to help train, test, and evaluate basic ML models under guidance.

Key Responsibilities

- Clean and preprocess datasets used for advertising and e-commerce analytics.
- Help train, test, and evaluate basic ML models under guidance.
- Run experiments, compare results, and summarize learnings.
- Write small Python scripts/notebooks for feature engineering and analysis.
- Document work clearly so it can be reused by the team.



www.createaigenie.com



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+91 74280234823

Bibliography/References

You can use the following references for your internship report:

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7. **Anthropic.** *Claude Documentation.* Anthropic.
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10. **Amitava Mondal.** *Internship Work and Project Documentation, Digital Hercules Innovations, 2026.*